goings we observe in the three dimensions of day-to-day life might themselves be holographic projections of physical processes taking place on a distant, two-dimensional surface. In their new and peculiar-sounding vision, we and everything we do or see would be akin to holographic images. Whereas Plato envisioned common perceptions as revealing a mere shadow of reality, the holographic principle concurs, but turns the metaphor on its head. The shadows—the things that are flattened out and hence live on a lower-dimensional surface—are real, while what seem to be the more richly structured, higher-dimensional entities (us; the world around us) are evanescent projections of the shadows.[45]

Again, while it is a fantastically strange idea, and one whose role in the final understanding of spacetime is far from clear, ’t Hooft and Susskind’s so-called *holographic principle* is well motivated. For, as we discussed in the last section, the maximum entropy that a region of space can contain scales with the area of its surface, not with the volume of its interior. It’s natural to guess, then, that the universe’s most fundamental ingredients, its most basic degrees of freedom—the entities that can carry the universe’s entropy much as the pages of *War and Peace* carry its entropy— would reside on a bounding surface and not in the universe’s interior. What we experience in the “volume” of the universe —in the *bulk*, as physicists often call it—would be determined by what takes place on the bounding surface, much as what we see in a holographic projection is determined by information encoded on a bounding piece of plastic. The laws of physics would act as the universe’s laser, illuminating the real processes of the cosmos—processes taking place on a thin, distant surface—and generating the holographic illusions of daily life.

We have not yet figured out how this holographic principle might be realized in the real world. One challenge is that in conventional descriptions the universe is imagined either to go on forever, or if not, to wrap back on itself like a sphere or a video game screen (as in Chapter 8), and hence it wouldn’t have any edges or boundaries. So, where would the supposed “bounding holographic surface” be located? Moreover, physical processes certainly seem to be under our control, right here, deep in the universe’s interior. It doesn’t seem that something on a hard-